

Analysis of Adaptive Mutation in Crazy Particle Swarm Optimization

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Abstract—This paper contributes an investigation in the utilization of adaptive mutation in a very prominent edition of particle swarm optimization technique i.e. crazy particle swarm optimization for the betterment of finding out the global best solution. In the field of swarm intelligence, particle swarm optimization is recognized as one of the popular techniques, which have shown an accepted performance in various real world engineering problems. Simultaneously, it has some cons which inspire the researchers for further modifications to get a better output with the improvement of its convergence quality. Here we proposed one enhanced version of Crazy PSO i.e. Adaptive mutated Crazy PSO. After the results of simulating experiment, it has been observed that Adaptive mutated Crazy PSO has exhibited improved results as compared to the basic Crazy PSO. The addition of Adaptive mutation helps in the achievement of better performance in Crazy PSO which has been tested on various benchmark functions

Keywords— Particle Swarm Optimization, Crazy PSO Algorithm, Adaptive mutated Crazy PSO Algorithm, Standard Benchmark functions.

I. INTRODUCTION

The engineering world needs several algorithms for solving any optimization problem. The process of finding out the proper solution requires the techniques for maximization or minimization of the objective function utilized to find the solution. In the process of optimization maximum return is expected with less resource utilization. Evolutionary algorithms are quite successful in case of solving numerous optimization problems. The concepts of swarm intelligence which basically deals with a number of populations those are having interaction with each other and their respective surroundings are the base of this algorithm. Particle Swarm Optimization (PSO) is one of the recognised swarm intelligence logical procedure developed by Kennedy and Eberhart in 1995 as the most efficient method for nonlinear problems. In PSO, each particle in the population has been assigned with a proper velocity which leads to have good control on the particles and acquires better convergence. It is one of the popular techniques which is applied on various fields such as neural network, Image processing, system identification etc. However, the basic PSO also has the

problem of acquiring the local optima and premature convergence while resolving complex problems. So, one modified version was proposed i.e., crazy PSO which is regarded as one of the widely used modified version by several researchers. This crazy PSO algorithm also utilized in several articles for different applications. Since, the area of machine learning is developing at a very faster rate, it needs new adaptive algorithms to handle the supervised learning in various complicated applications. This paper planned one further improvement feature to include in the crazy PSO algorithm for the refinement of this crazy PSO. It is done by applying adaptive mutation to produce enhanced superior results. Hence the modified approach has named as Adaptive mutated Crazy PSO (ACPSO). The modified version is giving finer results than the Crazy PSO when tested in various standard benchmark functions. This paper is discussing the performance of Crazy PSO as well as Adaptive mutated Crazy PSO under various simulating experiment conditions and the paper is arranged in order as follows. In a nutshell, section 2 describes about Crazy PSO and the recently proposed moderation i.e. Adaptive mutated Crazy PSO (ACPSO). Section 3 narrates detailed result analysis. At the end it's the conclusion which got narrated in section 4.

II. COMPUTATIONAL TECHNIQUES

A. Crazy PSO Algorithm(CPSO)

PSO algorithm [1] is based on the strategy of population and respective fitness value of every individual constituent of the population. Every particle present in the population has two important parameters i.e. position and velocity. After updating the velocity and position in each iteration, it deals with two values i.e. personal_best and global_best. These two values combining inspire the recent velocity of the particles to go towards a proper direction for optimal solution of the problem under consideration. So considering global_best as the optimal value, if there occurs a premature convergence then global_best will firmly taken the local optima value. From Eq. (1) if we modify the global_best then the direction of path of all the particles will be changed. Thus all the swarm particles will start searching through other locality. While performing the consecutive process,

the algorithm will find freshly generated optima i.e. personal best as well as global best. The improved modification of PSO algorithm which is referred as Crazy PSO algorithm is considered as one of the finest modifications of the basic PSO algorithm. This modified version shows superior output with higher convergence as compared to the basic PSO. Here the theory of craziness is presented with the moderation in velocities of particles for better exploration of the entire search space along with the exploitation of each point. Craziness probability is being introduced which will support the divergence of particles present in the entire population. This craziness probability provides a way to support for an enhanced exploration [2]. The expression of the normal PSO is not described as the mathematical relation of crazy PSO also contains the expression of standard PSO along with the extra crazy PSO modifications. In the domain of mathematics, the Crazy PSO algorithm [3] can be represented as

$$Velocity_i^{(t+1)} = x + y + z + P(k_4)Sign(k_4)v_d^{craziness} \quad (1)$$

$$x = k_2 * sign(k_3) * Velocity_i^t$$

$$y = (1 - k_2) * C_1 * k_1 * (particlebest_i^t - Y_i^t)$$

$$z = (1 - k_2) * C_2 * (1 - k_1) * (globalbest^t - Y_i^t)$$

$$Y_i^{t+1} = Y_i^t + Velocity_i^{t+1} \quad (2)$$

$$Sign(k_3) = \begin{cases} -1 & k_3 \leq 0.05 \\ 1 & k_3 > 0.05 \end{cases}$$

$$P(k_4) = \begin{cases} 1 & k_4 \leq P_{cr} \\ 0 & k_4 > P_{cr} \end{cases}$$

$$Sign(k_4) = \begin{cases} 1 & k_4 \geq 0.5 \\ -1 & k_4 < 0.5 \end{cases}$$

Where P_{cr} is a recognized as the predefined probability of craziness, k_1 , k_2 , k_3 and k_4 are the utilized arbitrary parameters normally selected homogeneously within the interval $[0, 1]$; $v_d^{craziness}$ is also an variable of arbitrary nature whose value is unvaryingly in the interval $[v_d^{\min}, v_d^{\max}]$ and the enhanced velocity and position in every iteration is given by Y_i^{t+1} , $Velocity_i^{(t+1)}$

B. Adaptive mutated Crazy PSO Algorithm(ACPSO)

The Crazy PSO is the algorithm which is providing a better investigation of the entire search space but the exploitation is not sufficient when deals with complicated engineering optimization problems. The Adaptive mutated crazy PSO algorithm embedded the feature of adaptive mutation [4] in the execution process of normal crazy PSO algorithm. This feature is an attempt to enhance the exploitation produced by the standard referred algorithm during the process of

simulation. This technique will help the particles to approach towards the global minimum without being trapped in different local minimum and this process generally helps in finding a more preferred solution to any optimization problems under consideration. The adaptive nature of mutation has the merit of not requiring a very large domain of space of exploration for the betterment of the solution to the problem under investigation.

According to the following equation the adaptation will continue among the particles at the end of iteration. The mathematical expression for adaptive mutation is given by

$$position_{ij} = position_{ij} + \sigma'_{ij} * betarand_j() \quad (3)$$

Where $\sigma'_{ij} = \sigma_{ij} * \exp(\tau NN(0,1) + \tau' NN_j(0,1))$

$NN(0,1)$ is giving a arbitrary value with normal distribution having mean 0 as well as standard deviation 1.

The τ and τ' are defined as $1/\sqrt{2n}$ and $1/\sqrt{2\sqrt{n}}$ respectively. Similarly, the $Betarand()$ generates arbitrary numerical value utilizing the beta distribution with parameters less than one.

The logical sequence of execution of projected algorithm is specified as below

- Initialize the population randomly.
- Initialize all the parameters utilized in the experiment.
- Evaluate the fitness function of the whole swarm population.
- Comparison will be done with the previous fitness value and freshly generated fitness value and the best one will be assigned as `global_best`.
- Updation of velocity will be done by adopting Eq.(1) and Eq.(2).
- Revise the `personal_best` as well as `global_best`.
- Apply the adaptive mutation.
- Replace the adaptive mutated value as the `global_best` value
- Updation of particle vectors will be done in each iteration.
- Iteration will last till it attains to the maximum iteration.

III. RESULT ANALYSIS

For testing the effectiveness as well as utility of the planned method, some standard engineering and real world benchmarking functions [5] are taken into consideration. The simulating experiment includes Crazy PSO as well as Adaptive mutated Crazy PSO algorithm. The Basic PSO algorithm is not included in the simulating experiment as the higher version of PSO i.e., Crazy PSO is acting as the reference. The reference papers already provide the superiority of the referred crazy PSO than the usual PSO algorithm. The results of this proposed method is in this paper compared with the outputs of already existed modified

technique i.e., Crazy PSO. Minimization of the output of benchmarking functions by simulation is the important criteria. Some of the standard parameters used in Crazy PSO are $C1=1$, $C2=4-C1$, Population = 100, dimension = 4. Both the Crazy PSO(CPSO) and Adaptive mutated Crazy PSO(ACPSO) has been run 40 trials to obtain the simulated values listed in various tables. The tables presented in this section includes the finest value, worst value as well as and the mean value. Sample simulation profiles of all the utilized standard benchmarking functions are also presented in different figures to exhibit the improvement in the projected research.

Benchmark functions

De Jong Function

$$f(vv) = vv_1^2 + vv_2^2 + vv_3^2$$

Range is $-2.56 \leq vv_i \leq 5.12$ with $i=1, 2, 3$.

Powell Function

$$f(vv) = \sum_{j=1}^{n/4} (vv_{4j-3} + 10vv_{4j-2})^2 + 5(vv_{4j-1} - vv_{4j})^2 + (vv_{4j-2} - 2vv_{4j-1})^4 + 10(vv_{4j-3} - vv_{4j})^4$$

Range is $-4 \leq vv_i \leq 5$ with $i=1, 2 \dots n$.

Levy function

$$f(vv) = \sin^2(\pi y_1) + \sum_{i=1}^{k-1} (y_i - 1)^2 (1 + \sin^2(\pi y_i + 1)) + (y_k - 1)^2 (1 + \sin^2(2\pi vv_k))$$

$$\text{Subject to } y'_i = 1 + \frac{vv_i - 1}{4}$$

Where $i=1, \dots, n$

Range is $-10 \leq vv_i \leq 10$ with $i=1, 2 \dots n$.

TABLE I. CRAZY PSO ALGORITHM

Functions	Results		
	<i>Best</i>	<i>Worst</i>	<i>Mean</i>
Funct_01	0.01768	0.01812	0.01774
Funct_02	0.06312	0.06826	0.06318
Funct_03	0.007201	0.007305	0.007202

TABLE II. ADAPTIVE MUTATED CRAZY PSO ALGORITHM

Functions	Results		
	<i>Best</i>	<i>Worst</i>	<i>Mean</i>
Funct_01	-0.05147	-0.05124	-0.05145
Funct_02	-0.00706	-0.00698	-0.007055
Funct_03	0.0009801	0.0009876	0.0009803

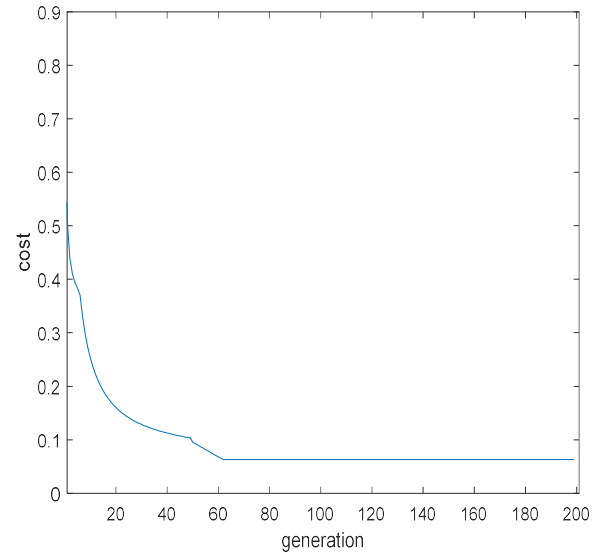


Fig. 1. Profile for Powell function using CPSO

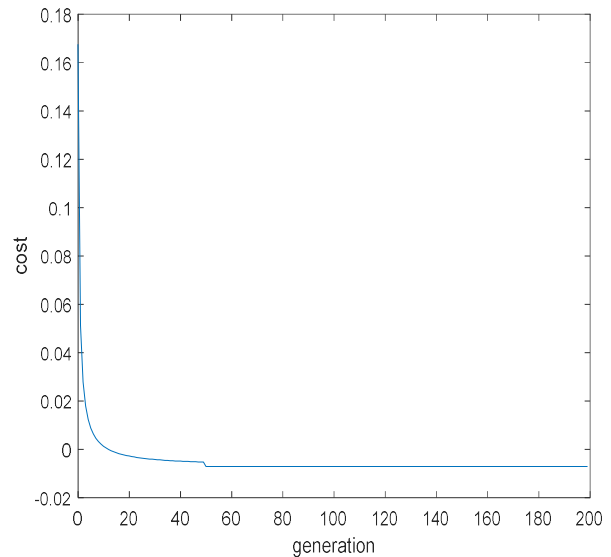


Fig. 2. Profile for Powell function using ACPSO

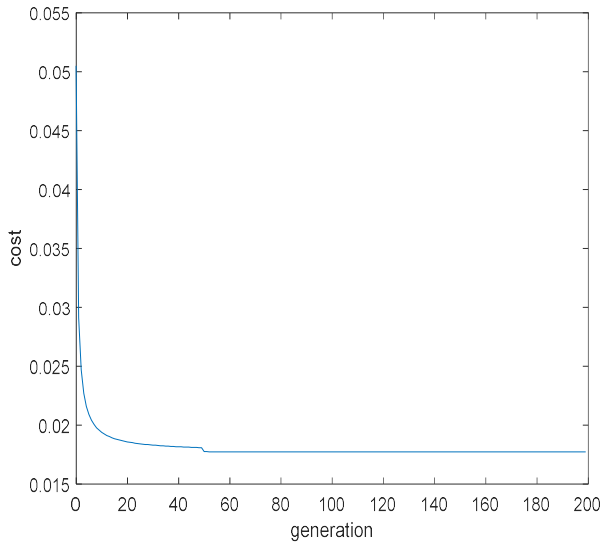


Fig. 3. Profile for de jounge function using CPSO

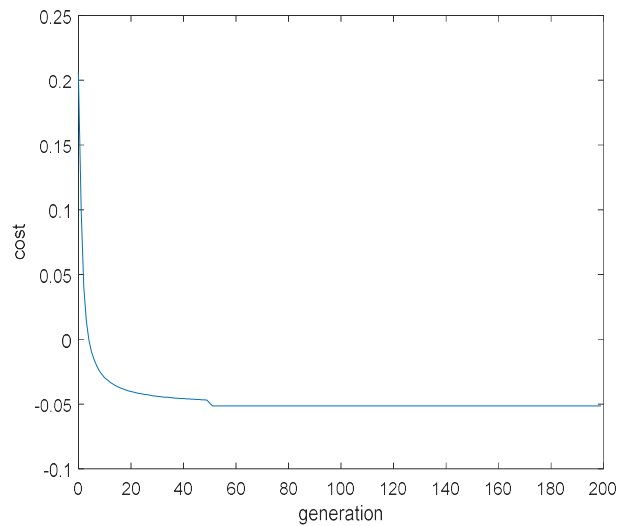


Fig. 4. Profile for de jounge function using ACPSO

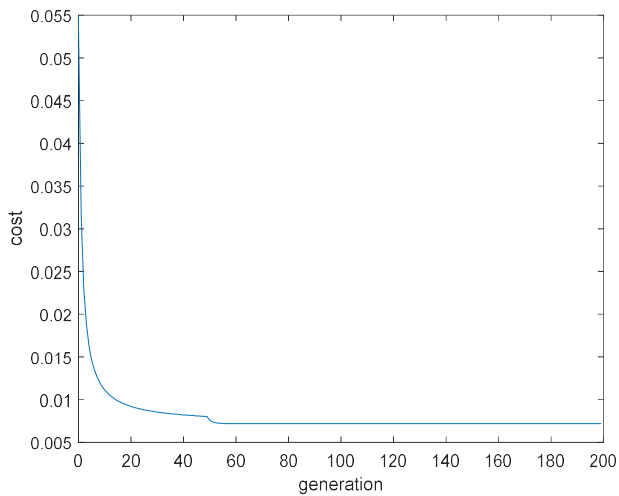


Fig. 5. Profile for Levy function using CPSO

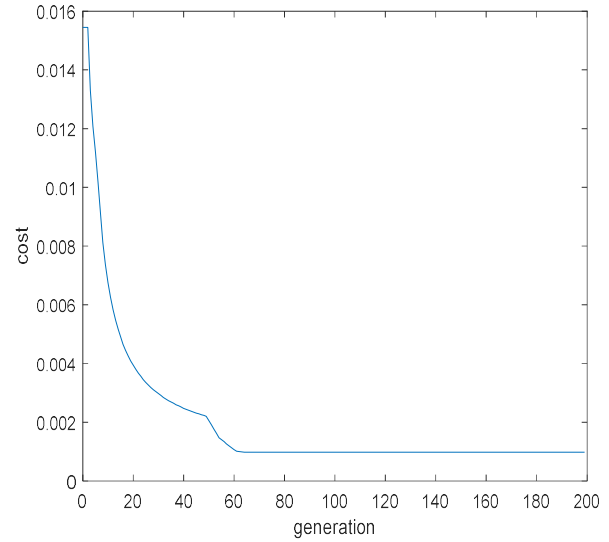


Fig. 6. Profile for Levy function using ACPSO

The results of the two algorithms i.e. Crazy PSO and Adaptive mutated Crazy PSO for the various simulated functions are presented in table I and II correspondingly. The tables are having best value, worst value and mean value of all the simulated experiments for the functions under examination. According to the values presented in the tables, it can be clearly stated that Adaptive mutated Crazy PSO gives superior results when the comparison is done with the basic crazy PSO. The results of PSO haven't taken into consideration as the moderations are prepared to the already existed customized versions of PSO i.e., Crazy PSO. For the De jounge function, the average value of Adaptive mutated Crazy PSO is -0.05145 which is clearly enhanced than crazy PSO i.e.0.01774. In the Powell function the mean values are 0.06318 and -0.007055 in Crazy PSO and Adaptive mutated Crazy PSO respectively. The mean values acquired for the Levy function in Crazy PSO and ACPSO are 0.007202 and 0.0009803 correspondingly. All the values of both the algorithms are clearly showing that Adaptive mutated Crazy PSO is proving its dominance in comparison to the Crazy PSO in every case of simulations. Similar conclusions can also be made if best values are taken into observation. In this case also, the Adaptive mutated Crazy PSO is also giving superior results for every function. Similar interpretation can be assigned for the performance of Adaptive mutated Crazy PSO if the worst values are taken into attention. Very important result can be derived for the values of best and worst for all the three involved functions for both the algorithms involved in the simulation. The separation between the best and worst values for function one for crazy PSO is higher than the crazy PSO with adaptive mutation. This phenomenon indicates that the precision of crazy PSO with adaptive mutation is higher in comparison to the referred crazy PSO algorithm. This precision is one of the most demandable requirements of any optimization algorithm for computational intelligence and machine learning applications. Similar conclusions can also be derived from the separation of best as well as worst values of remaining two functions for both the implemented algorithm. In both cases, the precision of crazy PSO with adaptive mutation is

dominating the standard crazy PSO algorithm. This precision of repeated measurements is a very important advantage of the newly modified algorithm.

The results analysis section also contains the six graphical plots for demonstrating the efficiency of the projected approach in fig 1 to 6 respectively. Fig 1 and 2 are for the function two, fig 3 and 4 are for the function one whereas the figure 5 and 6 are for the function 3 respectively. In all these plots, the Adaptive mutated Crazy PSO is exhibiting a better reduced cost while it is compared with the already existed edition of PSO i.e., Crazy PSO. Also, the Adaptive mutated Crazy PSO is providing a superior cost with reduced number of iterations. This feature of achievement of reduced cost value at the lower number of iterations in the newly projected modified version of algorithm in this paper saves a lot of time of computation for finding the solutions of complex engineering problems.

IV. CONCLUSION

In this paper, one enhanced modified version of Crazy PSO is recommended for providing a superior performance as compared to the already existed algorithm i.e. Crazy PSO. The recommended approach of freshly modified algorithm has named as Adaptive mutated Crazy PSO. The proposed modified and improved version of the Crazy PSO in this paper is tested with some standardized well known benchmarking functions. This testing with these functions proved the superiority of newly version of projected algorithm over the standard Crazy PSO algorithm. As already verified this algorithm will perform much better than the standardized PSO algorithm. The newly moderation method is also very simple to execute and also the complexity in newly moderated method is not so difficult to

deal with when will be implemented for various real life complex engineering problems. The proposed method comes up with a better local search as well as global search. So the proposed method can easily be applied in any practical engineering problems according to the appropriate requirement. Therefore, the Adaptive mutated Crazy PSO for its finer outputs can be better utilized to find the solutions to various complex modern engineering applications.

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